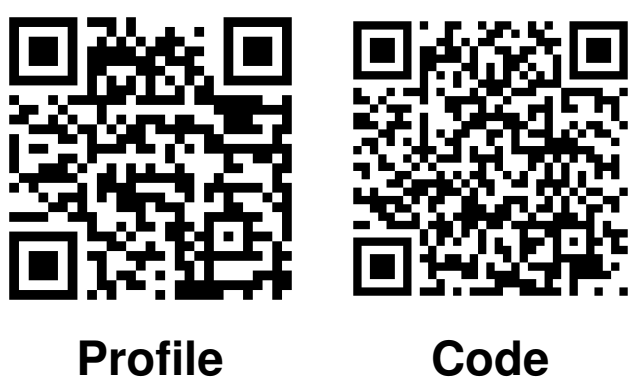


DeepNucleiNet: Encoding Spatial Nuclear Patterns for Data-Efficient Necrosis Detection in Computational Neuropathology

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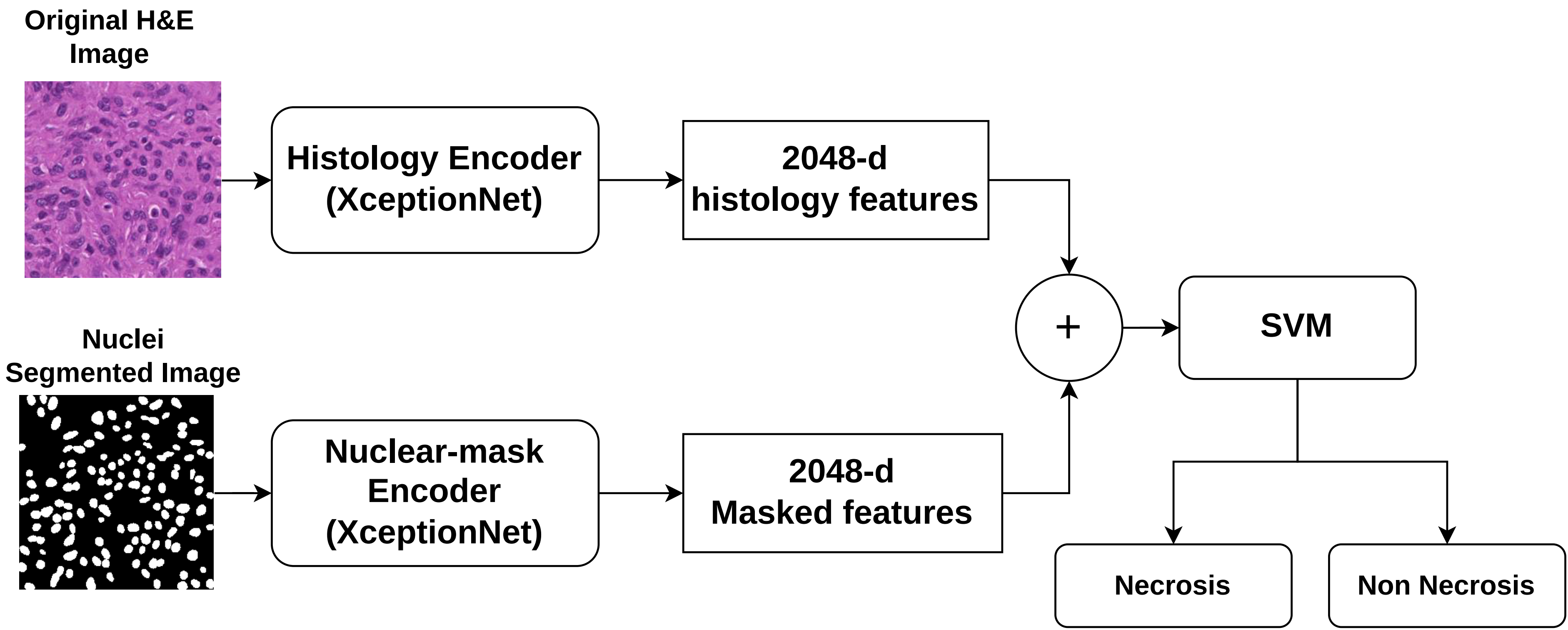


Problem

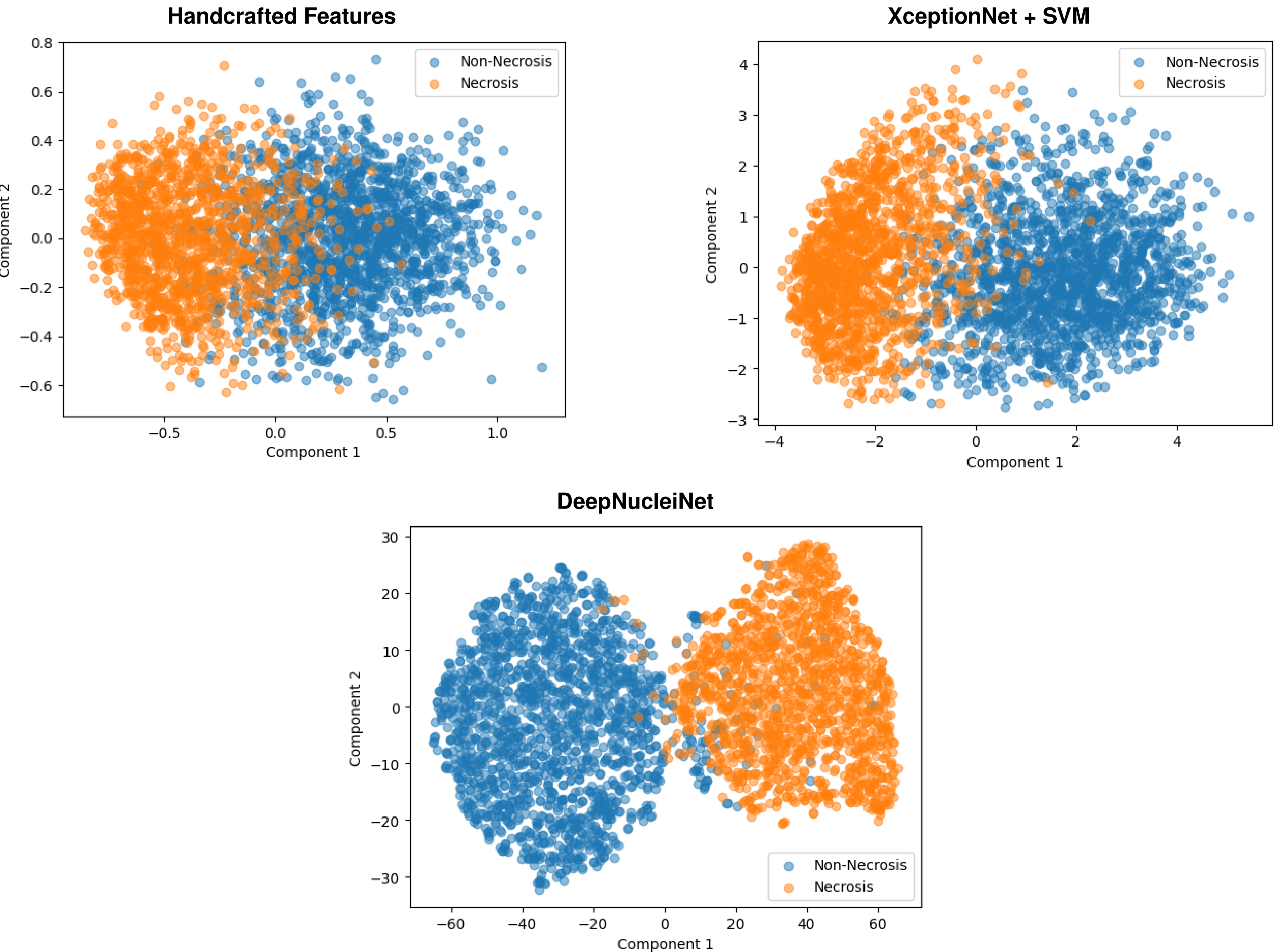
Necrosis is a clinically important histologic pattern marked by **loss of viable tumor cells, nuclear depletion**, and disrupted tissue architecture. Reliable detection is challenging because necrotic regions are morphologically heterogeneous, while training robust models requires substantial expert annotation.

- In small H&E tiles, ML and deep learning models may struggle to distinguish necrosis from acellular stroma, hemorrhagic or degenerated regions, and preparation-related artifacts.
- Appearance-only representations may miss the spatial organization and depletion of nuclei that are central to how necrosis is identified by pathologists.

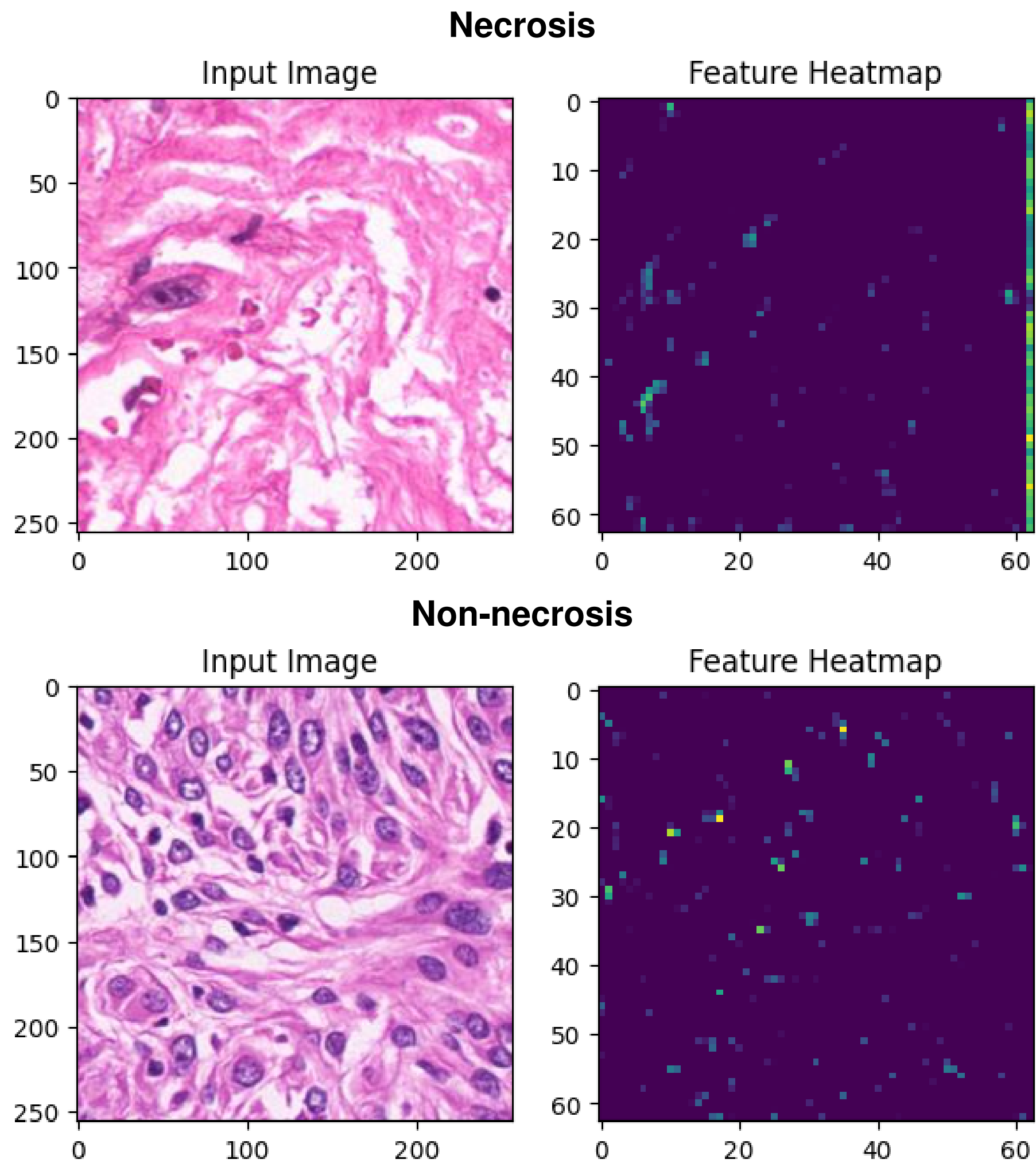
DeepNucleiNet Architecture



Feature Representation



Appearance-Based Feature Attention



Dataset and Evaluation

Meningioma H&E whole-slide images from NIMHANS with expert necrosis annotations: 10,233 tiles (4,040 necrosis; 6,193 non-necrosis).

Predictive Performance

Model	F1	Bal. Acc	MCC	Sens.
Baseline (SVM)	0.324	0.445	-0.073	0.395
Nuclei Density + LR	0.593	0.785	0.428	0.681
XceptionNet (FC)	0.674	0.791	0.436	0.800
Virchow2 + MLP	0.730	0.737	0.438	0.759
XceptionNet (H&E)	0.847	0.893	0.681	0.842
UNI2-h + MLP	0.884	0.913	0.742	0.901
XceptionNet (Nuclei)	0.904	0.911	0.752	0.901
DeepNucleiNet	0.942	0.936	0.820	0.945

- Combining tissue appearance with explicit nuclear structure provided the most consistent discrimination among the evaluated representations.

Cross-Tumor Generalization

Dataset	Domain	F1	MCC
TCGA GBM	Grade 4 glioma	0.943	0.841
DeepHisto	Mixed glioma	0.843	0.741
TiGER	Breast TNBC	0.635	0.382
TCGA LGG	Low-grade glioma	—	—

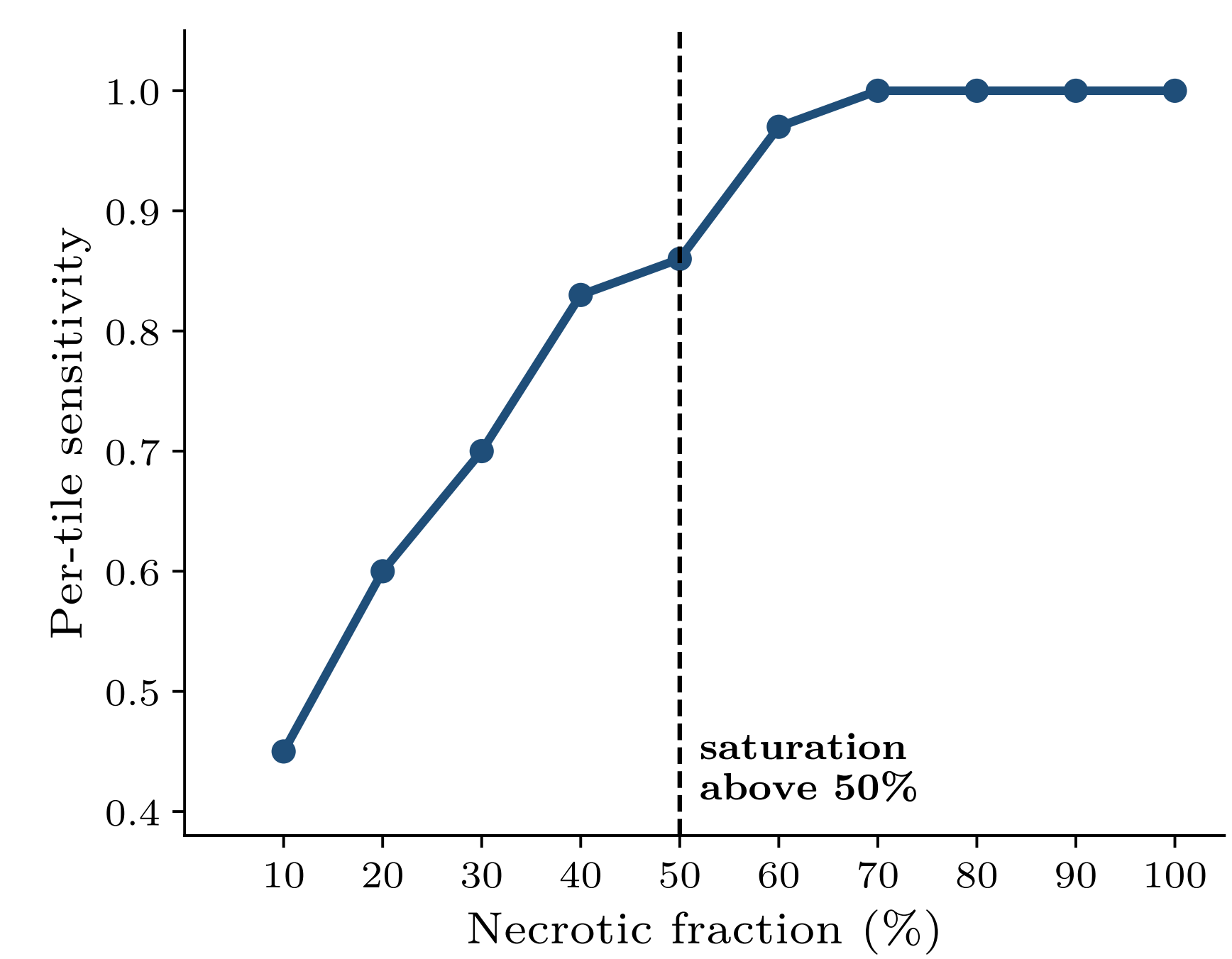
- Transfer performance was higher in related CNS tumors and decreased with increasing morphologic distance from the training domain.

Grid-Position Analysis

Grid origin	Tiles	Necrosis	Non-necrosis	Performance
X	3162	3014	148	0.9539
X+100	3160	3012	148	0.9600
X+200	3164	3015	149	0.9750
X+300	3165	3015	151	0.9500
X+400	3167	3017	150	0.9520

- Performance remained stable across grid offsets, indicating limited sensitivity to tile-origin displacement.

Sensitivity to Necrotic Fraction



Representative Morphology and Nuclear Patterns

